LAB MANUAL

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (AI & ML)

Database & Management System

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Experiment-1

Aim

To design an Entity–Relationship (ER) diagram for a company database system and define the relationship types between entities.

Objectives

- To identify entities and their attributes in a company database
- To represent relationships among entities
- To define relationship types such as one-to-one, one-to-many, and many-to-many

Entities and Attributes

1. Employee

Represents employees working in the company.

Attributes:

- SSN (Primary Key)
- Name
- Address
- Sex
- Salary
- SuperSSN
- DNo

2. Department

Represents departments in the company.

Attributes:

- DNo (Primary Key)
- DName
- MgrSSN
- MgrStartDate

3. Project

Represents projects handled by the company.

Attributes:

- PNo (Primary Key)
- PName
- PLocation
- DNo

4. DLocation

Represents locations of departments.

Attributes:

- DLoc
- DNo

5. Works_On

Represents the association between employees and projects.

Attributes:

- SSN
- PNo
- Hours

<u>Relationships and Relationship Types</u>

1. Works_In

Entities involved: Employee – Department

Relationship type:

- **Many-to-One (N : 1)**

Explanation:
Many employees can work in one department, but each employee works in only one department.

2. Manages

Entities involved: Employee – Department

Relationship type:

- **One-to-One (1 : 1)**

Explanation:
Each department is managed by one employee, and an employee manages at most one department.

3. Has_Location

Entities involved: Department – DLocation

Relationship type:

- **One-to-Many (1 : N)**

Explanation:
A department can have multiple locations, but each location belongs to only one department.

4. Controls / Involves

Entities involved: Department – Project

Relationship type:

- **One-to-Many (1 : N)**

Explanation:
A department can control multiple projects, but each project is controlled by only one department.

5. Works_On

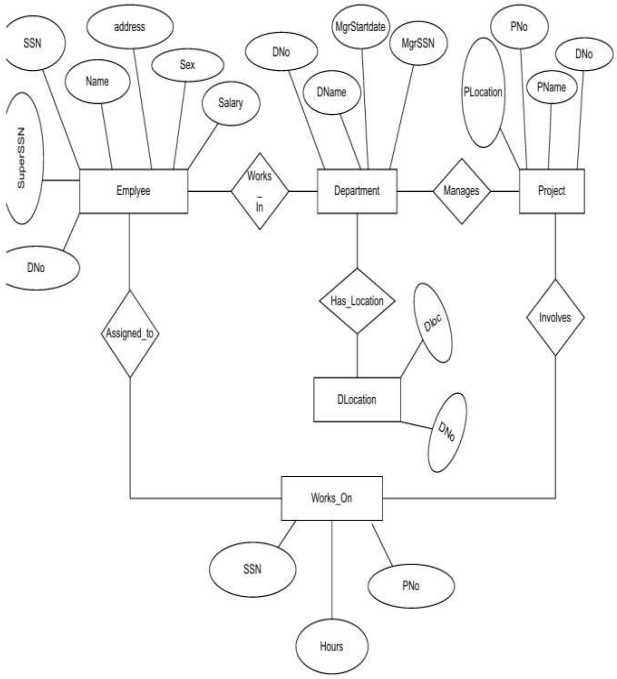
Entities involved: Employee – Project

Relationship type:

- **Many-to-Many (M : N)**

Explanation:
An employee can work on multiple projects, and a project can have multiple employees.
The attribute **Hours** indicates the number of hours an employee works on a project.

Result:


SSN
address
Name
Sex
Salary
SuperSSN
DNo
Emplyee
Works
_
In
DNo
DName
MgrStartdate
MgrSSN
Department
Manages
PNo
PLocation
PName
DNo
Project
Assigned_to
Has_Location
DLoc
DLocation
DNo
Involves
Works_On
SSN
PNo
Hours